

# Thermal Properties of Matter

## # Temperature and Heat

**Heat** :: Definition: Heat is a form of energy that flows from one body to another due to a temperature difference

- ∴ Symbol:  $Q$
- ∴ SI unit: Joule
- ∴ another units: calorie (cal)
- ∴ Direction of flow: Always flows from a body at higher temperature to one at lower temperature
- ∴ Nature: Heat is energy in transit, not contained in a body.

## Temperature:

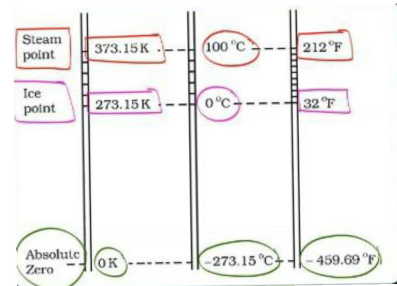
- ∴ Definition: Temperature is the measure of the degree of hotness or coldness of a body
  - ∴ It indicates the direction of flow of heat
  - ∴ physical meaning: It is proportional to the average kinetic energy of the molecules of a substance
- Average kinetic energy  $\propto T$

## # Measurement of Temperature

| Scale                             | Freezing Point of Water | Boiling Point of Water | Interval Division | Relation                |
|-----------------------------------|-------------------------|------------------------|-------------------|-------------------------|
| Celsius ( $^{\circ}\text{C}$ )    | $0^{\circ}\text{C}$     | $100^{\circ}\text{C}$  | 100 equal parts   | --                      |
| Fahrenheit ( $^{\circ}\text{F}$ ) | $32^{\circ}\text{F}$    | $212^{\circ}\text{F}$  | 180 equal parts   | $F = \frac{9}{5}C + 32$ |
| Kelvin (K)                        | $273.15\text{ K}$       | $373.15\text{ K}$      | 100 equal parts   | $K = C + 273.15$        |

ex.  $75^{\circ}\text{C} \rightarrow \text{F}$   
 $F = \frac{9}{5} \times (75) + 35$   
 $F = 167^{\circ}\text{C}$

ex.  $50^{\circ}\text{C} \rightarrow \text{K}$   
 $K = (50^{\circ}\text{C}) + 273.15$   
 $= 323.15\text{ K}$



## # Ideal Gas Equation:-

★ Boyle's Equation

$$P \propto \frac{1}{V} \text{ (at constant temp)}$$

$$\Rightarrow PV = \text{constant}$$

★ Charles's law —

$$V \propto T \text{ (At constant pressure)}$$

★ Gay-Lussac's law

$$P \propto T \text{ (At constant volume)}$$

Merging these three equations we get:-

$$PV \propto T$$

$$\Rightarrow PV = MRT \rightarrow \text{Ideal gas Equation}$$

$P$  = pressure of the gas

$V$  = volume of the container

$T$  = Temperature (always in K)

$M$  = No of moles of the given gas sample

$R$  = universal gas constant

$$\Rightarrow R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$$

Pressure ( $P$ ), volume ( $V$ ) और Temperature ( $T$ ) में से एक quantity constant होनी ही चाहिए, सभी यह eqn follow होगी।

## # Absolute temperature

∴ The temperature measured from absolute zero ( $-273.15^{\circ}\text{C}$ )

∴ At absolute zero, molecular motion ceases and the gas pressure becomes zero.

∴ Therefore, the Kelvin scale is an absolute temperature scale

$$T(K) = t(^{\circ}C) + 273.15$$

## # Thermal Expansion



$$l_f = l_i (1 + \alpha \Delta T) \quad A_f = A_i (1 + \beta \Delta T) \quad V_f = V_i (1 + \gamma \Delta T)$$

$\alpha$  = coefficient of linear expansion  
 $\beta$  = coefficient of superficial expansion,  $\beta = 2\alpha$   
 $\gamma$  = coefficient of volumetric expansion,  $\gamma = 3\alpha$

$$\Delta T = \text{change in temperature} = (\text{Final temp} - \text{initial temp})$$

Gases > Liquids > Solids  
 ← Rate of Thermal Expansion

$$\alpha : \beta : \gamma = 1 : 2 : 3 \quad \begin{cases} \beta = 2\alpha \\ \gamma = 3\alpha \end{cases}$$

## # Thermal Expansion

Anomalous Expansion of water

∴ water contracts instead of expanding between  $0^{\circ}C$  and  $4^{\circ}C$

∴ volume is minimum at  $4^{\circ}C$

∴ This property helps aquatic life survive in cold regions as the water below remains at  $4^{\circ}C$  while ice forms on the surface

Applications of thermal Expansion

- 1 Gaps are left between railway tracks and bridges
- 2 overhead electrical wires are given slack
- 3 Bimetallic strips are used in thermostats and switches.
- 4 Glass vessels are made of pyrex glass to resist cracking

## # Heat Capacity

The heat capacity of a body is the amount of heat required to raise its temperature by  $1^{\circ}C$  (or  $1K$ )

$$S = \frac{\Delta Q}{\Delta T}$$

SI unit:

$$\text{Joule (Kelvin)}^{-1}$$

$$= JK^{-1}$$

Dimensional Formula

$$[ML^2T^{-2}K^{-1}]$$

Explanation: If a body has large heat to raise its temperature for example, water has a very high heat capacity compared to metals.

## # Specific Heat capacity

The specific heat capacity of a substance is the amount of heat required to raise the temperature of  $1kg$  of the substance by  $1^{\circ}C$  (or  $1K$ )

SI unit

$$Joule (Kelvin)^{-1} kg^{-1}$$

$$= JKg^{-1}K^{-1}$$

Dimensional Formula

$$[M^0L^2T^{-2}K^{-1}]$$

Formula for Heat Transfer

$$Q_{\text{absorbed or released}} = \Delta Q = ms\Delta T$$

## Explanation

If  $4200J$  of heat is supplied to  $1kg$  of water and its temperature rises by  $1K$ , then the specific heat capacity of water is.

$$C = \frac{Q}{m\Delta T} = \frac{4200}{1 \times 1} = 4200 JKg^{-1}K^{-1}$$

## Molar Specific Heat capacity

The molar specific heat capacity of a substance is the amount of heat required to raise the temperature of  $1$  mole of the substance by  $1^{\circ}C$  (or  $1K$ )

$$SI \text{ unit} = JK^{-1}mol^{-1}$$

Dimension formula =  $[ML^2T^{-2}K^{-1}mol^{-1}]$

$$C = \frac{S}{M} = \frac{1}{M} \frac{\Delta Q}{\Delta T}$$

no of moles of the given gas sample

→ Molar specific heat capacity at constant pressure  $\rightarrow C_p$

→ Molar specific heat capacity at constant volume  $\rightarrow C_v$

\*  $C_p - C_v = R$  — Mayer's formula  
 $\rightarrow R = \text{universal gas constant}$

\*  $s = \frac{C}{M}$  — Molar specific heat capacity  
 $\downarrow$   
 Molar mass

Specific heat capacity

\* Process with constant Temperature  
 → Isothermal process

\* Process with constant pressure  
 → Isobaric process

\* Process with constant volume  
 → Isochoric process

## # Calorimetry

∴ When two bodies at different temperature are placed in contact, heat flows from the hot body to the cold body until both attain the same final temperature.

∴ The device used to measure the quantity of heat gained or lost by a body is called a calorimeter.

ex (Hot) ball  $\rightarrow$  T  $\rightarrow$  copper calorimeter (water) (cold)

$$Q = ms\Delta T \quad Q_{\text{release}} = Q_{\text{absorbed}}$$

$$\Rightarrow m_1 s_1 (T_1 - T) = m_2 s_2 (T - T_2) + m_{H_2O} s_{H_2O} (T - T_2)$$

## # Principle of calorimetry

∴ The working of a calorimeter is based on the principle of conservation of Heat (Energy)

$$\text{Heat lost} = \text{Heat gained}$$

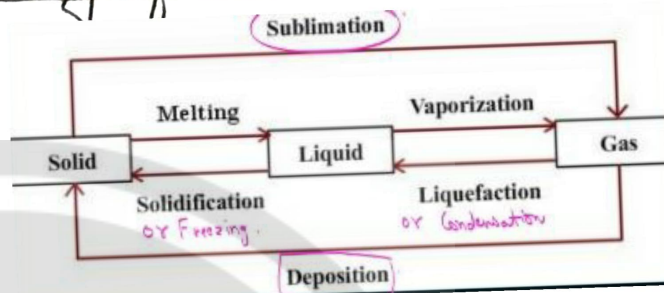
∴ That is, when a hot body is mixed with a cold body (or water)

inside a calorimeter, the heat lost by the hot body = heat gained by the cold body + calorimeter

$$Q_{\text{lost}} = Q_{\text{gained}}$$

∴ No heat is lost to the surrounding (assuming a perfectly insulated system)

## Change of state



## Latent Heat

∴ When a substance changes its state (solid  $\rightarrow$  liquid, liquid  $\rightarrow$  gas) at a constant temperature, heat is absorbed or released without any changes in temperature.

∴ This heat is called Latent Heat.

∴ "Latent" means hidden — This heat does not cause in temperature, but changes the state of matter



$$\text{Heat absorbed or released} = (\text{mass})(\text{latent heat})$$

$$Q = m h$$

sg unit: JOULE (J)      kg      sg unit:  $J kg^{-1}$

Dimension:  $[M^0 L^2 T^{-2}]$   
 formula

## Types of Latent Heat

### 1 Latent Heat of Fusion ( $L_f$ )

∴ Heat required to convert 1 kg of a solid into liquid at its melting point without change

in temperature

∴ Example Ice  $\rightarrow$  water at  $0^\circ\text{C}$

∴ unit  $\text{J/kg}$

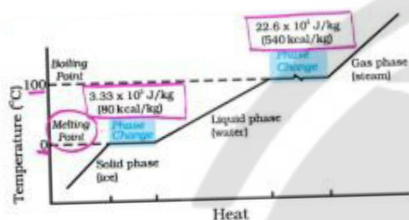
∴ For Ice  $L_f: 3.34 \times 10^5 \text{ J/kg}$

## 2 Latent Heat of vaporization ( $L_v$ )

∴ Heat required to convert  $1 \text{ kg}$  of liquid into vapour at its boiling point without change in temperature

∴ Example - water  $\rightarrow$  steam at  $100^\circ\text{C}$

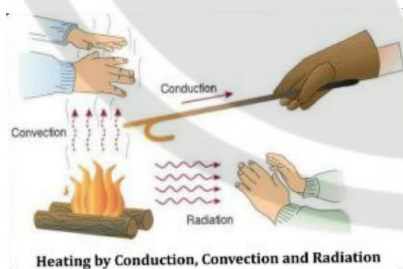
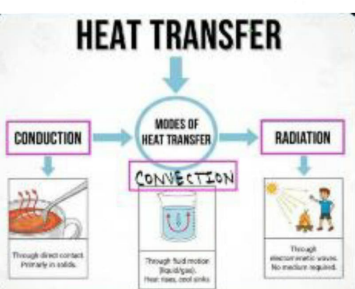
∴ For Ice  $L_v: 2.26 \times 10^6 \text{ J/kg}$



Temperature versus Heat for water  
1 atm pressure (not to scale)

## # Heat Transfer

The process by which heat energy flow from a hotter body to a colder body is called heat transfer.



## # Conduction

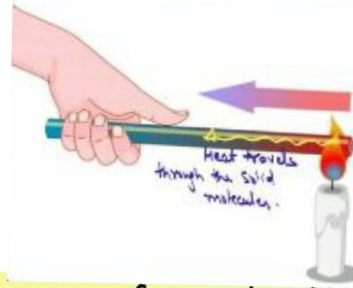
Conduction is the transfer of heat through a material without the movement of the material itself.

∴ It mainly occurs in solids, especially metals.

∴ Heat flows due to collision of

molecules or free electrons.

∴ Example: Heating one end of a metal rod - the other end becomes hot after some time.



## Law of Thermal conduction

$$H = \frac{Q}{t} = \frac{KA(T_1 - T_2)}{l}$$

$\rightarrow H$ : Heat transferred per unit time or Heat current

$\rightarrow Q$ : Heat Transferred

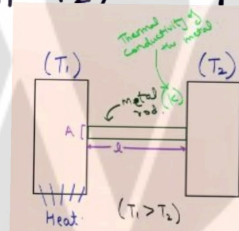
$\rightarrow t$ : given time

$\rightarrow K$ : Thermal conductivity of the material

$\rightarrow A$ : Area of cross-section of the material rod.

$\rightarrow l$ : length of the rod

$\rightarrow (T_1 - T_2)$ : Temperature difference



## Thermal conductivity ( $K$ )

∴ It is a measure of the material's ability to conduct heat.

∴ unit:  $\text{W/m}\cdot\text{K}$  or  $\text{J/s}\cdot\text{m}\cdot\text{K}^{-1}$   
( $\text{Watt}/(\text{meter}\cdot\text{Kelvin})$ )

∴ Good conductors: silver, copper, Aluminium] value of  $K$  is more

∴ Bad conductors / Insulators: wood, Glass, Plastic]  $\rightarrow$  value of  $K$  is less

## # convection

convection is the transfer of heat by actual movement of the particles of a fluid (liquid or gas). occurs only in liquids and gases, not in solids.

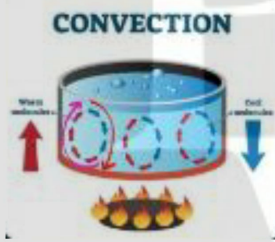
- ∴ When a fluid is heated, the hot part expands, becomes less dense, and rises.
- ∴ The cooler, denser fluid moves down to take its place.
- ∴ This circulation forms convection currents.

### Examples:

- ∴ Boiling water in a pot
- ∴ Sea and land breeze
- ∴ Heating rooms by a heater

### Applications:

- ∴ Radiators in vehicles and buildings use convection to distribute heat.
- ∴ chimneys use convection to remove hot air.

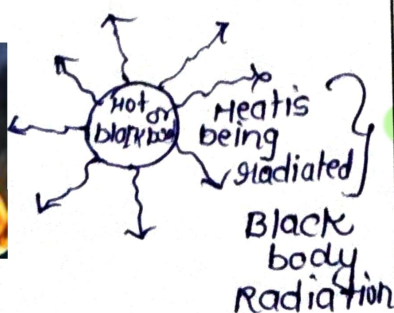


## # Radiation

Radiation is the transfer of heat in the form of electromagnetic waves. Infrared radiations without any medium. It can occur through vacuum.

### Example

- ∴ Heat from the sun reaches Earth through radiation.
- ∴ Feeling warmth near a fire.



## # Characteristics of Thermal Radiation:

- ∴ Travels at the speed of light
- ∴ Does not require a medium
- ∴ can be reflected, absorbed, or transmitted.
- ∴ Black bodies are good absorbers and emitters
- ∴ Shiny surfaces are poor absorbers but good reflectors.

## Laws related to Radiation

### ① Wein's displacement law:-

$$\lambda_m \propto \frac{1}{T}$$

$$\Rightarrow \lambda_m T = \text{constant}$$

$$\Rightarrow \lambda_m T = b \quad \text{Wein's displacement constant}$$

wavelength of the radiated wave at max intensity (meters m)  $\lambda_m$   
Absolute Temperature (K)  $T$   
( $b = 2.89 \times 10^{-3} \text{ mK}$ )

### ② Stefan-Boltzmann's law

$$E = e \sigma A T^4$$

Energy or Heat Radiated per unit time  $E$   
emissivity or (dimensionless)  $e$   
Stefan Boltzmann constant  $\sigma$   
Cross-section area  $A$   
Absolute Temp  $T$

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

$$E = e \sigma A (T - T_g)$$

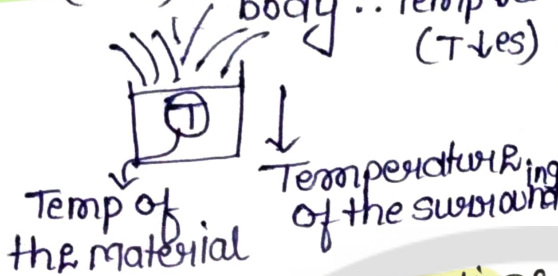
Absolute temp of the surrounding  $T_g$

## # Cooling: Newton's law of cooling

The rate of cooling (loss of heat per unit of a body is directly proportional to

The differences of temperature in between the body and the surround provided the temperature differences is small

$\frac{dQ}{dt} \propto (T - T_s)$  Here, the -ve sign indicates that is lost from the body  $\therefore$  Temp  $\downarrow$ es ( $T \downarrow$ es)



**Derivation of Temperature-time Relation of Newton's law of cooling:-**  
We know that:-

$Q = ms \Delta T$  or  $\Delta Q = ms \Delta T$   
Dividing both sides by time difference ( $\Delta t$ )

$$\Rightarrow \frac{\Delta Q}{\Delta t} = ms \frac{\Delta T}{\Delta t} \Rightarrow \left[ \frac{dQ}{dt} = ms \frac{dT}{dt} \right] \text{--- (2)}$$

Putting eqn (2) in eqn (1),

$$\Rightarrow ms \frac{dT}{dt} = -K(T - T_s)$$

$$= \int_{T=T_0}^{T=T} \frac{dT}{(T - T_s)} = \frac{-K}{ms} \int_{t=0}^{t=t}$$

[Let the temperature of the body at time  $t=0$  is  $T=T_0$ ]

$$\Rightarrow [\ln(T - T_s)]_{T_0}^T = \frac{-K}{ms} [t]_{t=0}^{t=t}$$

$$\Rightarrow \ln(T - T_s) - \ln(T_0 - T_s) = \frac{-K}{ms} (t - 0)$$

$$\Rightarrow \log_e \left( \frac{T - T_s}{T_0 - T_s} \right) = \frac{-K}{ms} t \quad \left[ \ln A - \ln B = \ln \left( \frac{A}{B} \right) \right]$$

$$\Rightarrow \frac{T - T_s}{T_0 - T_s} = e^{\frac{-K}{ms} t} \quad \left[ \log_e x = y \Rightarrow x = e^y \right]$$

$$\Rightarrow T = T_s + (T_0 - T_s) e^{\frac{-K}{ms} t}$$

Temp of the body  $\downarrow$  surrounding temp  $\uparrow$  body temp at  $t=0$

Some imp reason

$$ms \frac{dT}{dt} = -K(T - T_s)$$

$$\Rightarrow \frac{dT}{dt} = \frac{-K}{ms} (T - T_s)$$

$$\text{Let } \frac{-K}{ms} = K;$$

$$\Rightarrow \boxed{\frac{dT}{dt} = K(T - T_s)} \text{--- (i)}$$

$$\ln(T - T_s) - \ln(T_0 - T_s) = \frac{-K}{ms} (t - 0)$$

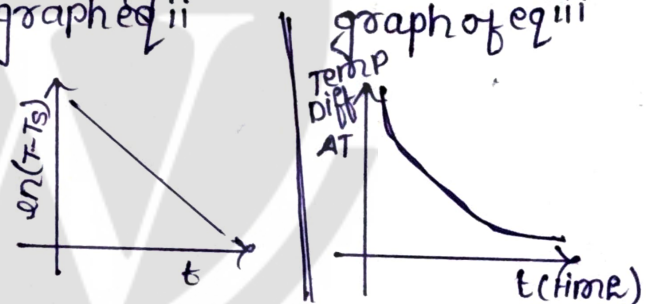
$$\Rightarrow \underbrace{\ln(T - T_s)}_y = \underbrace{\left( \frac{-K}{ms} \right)}_{\text{Slope (m) (-ve)}} t + \underbrace{\ln(T_0 - T_s)}_{\text{Constant}} \text{--- straight line}$$

$y = mx + c$

$$T = T_s + (T_0 - T_s) e^{\frac{-K}{ms} t}$$

$$\Rightarrow \boxed{T - T_s = (T_0 - T_s) e^{\frac{-K}{ms} t}} \text{--- (iii)}$$

$\Delta T$   $\downarrow$  change in temperature  
graphes ii  $\rightarrow$  exponential graph



**Newton's law of cooling**  
Conditions for validity  
Newton's law of cooling is valid only when?

$\therefore$  Temperature difference ( $T - T_s$ ) is small (say  $30^\circ\text{C}$ )

$\therefore$  The nature of the surface and medium remains constant

$\therefore$  Heat transfer is mainly through convection or radiation, not conduction.

## 2 Practical Applications

- ∴ cooling of hot liquids (like tea or coffee in a cup).
- ∴ Determining the time since death (in forensic science) using the cooling rate of a dead body.
- ∴ Designing radiators and cooling systems (car engines, electronics, etc)
- ∴ Heat treatment in industries controlling cooling rates of metals.